The OR joint-update filter

tracking carbon, disturbances and process error together

Claude

November 12, 2026

What the filter estimates

Each year t, at one site, from observations of wood carbon y_t:

- the carbon state $x_t = (\text{leaf},\ \text{soil},\ \text{wood})$
- whether a disturbance happened, and which kind: $z_t \in \{0, 1, \ldots, K\}$
- how large the model's process error is: precision τ

All three are uncertain and depend on each other, so they are updated jointly:

- a big drop in y_t is either a disturbance or a large process error,
- and the process error decides how far a forecast can move without a disturbance.

Model: one year

$$x_t = \begin{cases} f(x_{t-1}) + e_t & z_t = 0 \quad \text{(undisturbed)} \\ d_k\big(f(x_{t-1}) + e_t\big) & z_t = k \quad \text{(disturbance type } k\text{)} \end{cases} \qquad y_t = x_{t,\text{wood}} + \varepsilon_t$$

- f: one year of the VSEM forest model, driven by that year's light (PAR)
- $e_t \sim \mathcal{N}(0, Q)$, $\quad Q = \text{diag}\left(\sigma^2_{\text{leaf}},\ \sigma^2_{\text{soil}},\ 1/\tau\right)$ (leaf, soil: calibrated; wood: learned)
- $\varepsilon_t \sim \mathcal{N}(0, R_t)$, $\quad R_t = a\, y_t + b$ (error grows with biomass)
- $P(z_t = k) = \beta_{t,k}$: background disturbance rates, or the disturbance product's accuracy in the years it flags a disturbance

Model: disturbances and their process error

Disturbance k keeps a random *fraction* of leaf and wood; part of the removed carbon goes to the soil:

$$\text{leaf}', \text{wood}' \sim \mathcal{N}\big(\mu_{0,k} \cdot \text{old},\ V_{0,k} \cdot \text{old}^2\big), \qquad \text{soil}' = \text{soil} + \alpha\,(\text{removed})$$

$K = 16$ types: other, cut, fire, biotic, each split into size clusters fitted to observed biomass losses.

The year's process error happens before the disturbance, so it is scaled by the same fraction:

$$\text{Var}(\text{process error} \mid z_t = k) \;=\; D_k\, Q\, D_k, \qquad D_k = \text{diag}\left(\mu_{0,k}^{\text{leaf}},\ 1,\ \mu_{0,k}^{\text{wood}}\right)$$

Leaf and wood keep μ_0^2 of their process variance; the soil keeps all of it.

The posterior the filter carries

For every value τ_g on a grid ($g = 1..200$, log-spaced, process sd 0.01–3):

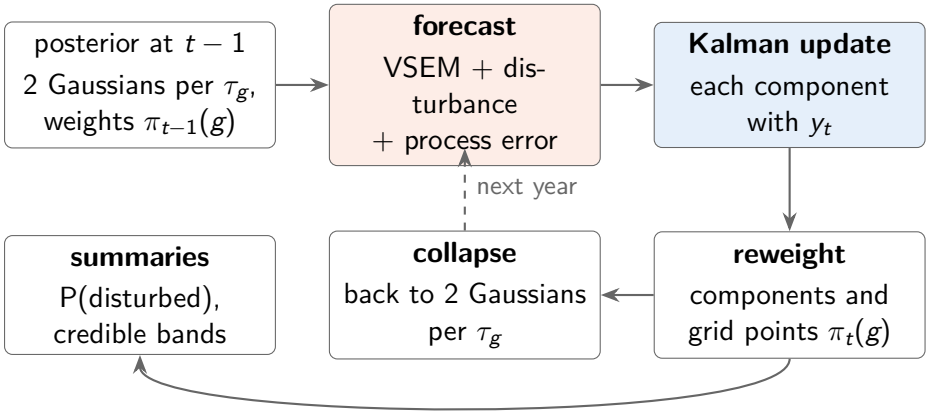
$$p(x_t \mid \tau_g, y_{1:t}) \;\approx\; \sum_j w_{g,j}\, \mathcal{N}(x_t;\; m_{g,j},\; C_{g,j}) \qquad j \in \{\text{undisturbed},\ \text{disturbed}\}$$

and a weight for each grid point:

$$\pi_t(g) \;=\; P(\tau = \tau_g \mid y_{1:t})$$

So the full posterior is a mixture of 2×200 Gaussians. Every summary (state, disturbance, τ) averages over it.

One assimilation step



Each grid point τ_g runs its own small mixture filter; they share the forecast simulations.

Forecast: push the posterior through VSEM

VSEM is nonlinear, so the forecast moments are estimated by **importance sampling**:

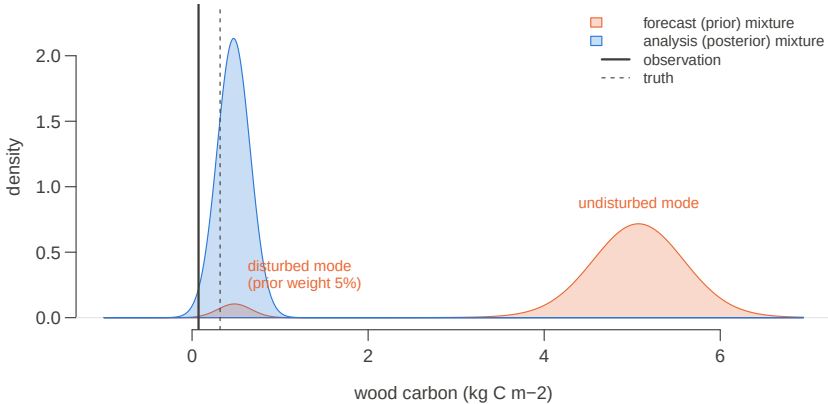
1. draw $N = 100$ states $x^{(n)}$ from the whole previous mixture (all g, j)
2. run each through one year of VSEM: $f(x^{(n)})$ (the expensive part: done once)
3. for each component (g, j), weight the samples by how likely it made them:

$$W_{g,j}^{(n)} \propto \frac{\mathcal{N}\left(x^{(n)};\ m_{g,j}, C_{g,j}\right)}{q\left(x^{(n)}\right)} \qquad q = \text{the mixture the samples came from}$$

4. for each disturbance type k: apply d_k to the simulated states, take the weighted mean and covariance, and add the process error (Q or $D_k Q D_k$, with $1/\tau_g$ for wood)

Result: forecast components (g, j, k) with means $\hat{m}$, covariances $\hat{C}$ and prior weights $w_{g,j}\,\beta_{t,k}$.

Forecast and analysis in a disturbance year



Synthetic run, year 10 (a harvest). The forecast mixture has an undisturbed mode and a disturbed mode near zero. The observation picks the disturbed one.

Kalman update: each component separately

Within a forecast component the model is linear-Gaussian, so the exact update applies (observing wood, $H = (0, 0, 1)$):

$$s = \hat{C}_{33} + R_t, \qquad K = \hat{C}_{\cdot 3}/s, \qquad m^a = \hat{m} + K(y_t - \hat{m}_3), \qquad C^a = \hat{C} - K\,\hat{C}_{3\cdot}$$

Each component also gets the likelihood of the observation, its predictive density:

$$L_{g,j,k} \;=\; \mathcal{N}\big(y_t;\ \hat{m}_3,\ s\big)$$

A component that predicted y_t badly keeps its update but loses its weight.

Reweighting: components and the process-error grid

Components, within each grid point (Bayes' rule over j, k):

$$w^{a}_{g,j,k} \;\propto\; w_{g,j}\,\beta_{t,k}\,L_{g,j,k}$$

Grid points: each one's evidence is how well its whole forecast predicted y_t:

$$\pi_t(g) \;\propto\; \pi_{t-1}(g)\;\underbrace{\sum_{j,k} w_{g,j}\,\beta_{t,k}\,L_{g,j,k}}_{p(y_t|\tau_g,\; y_{1:t-1})}$$

- $\pi_0(g)$: prior mass of the grid cell around τ_g
- small τ (large process error) explains jumps but spreads its forecast thin; large τ forecasts sharply but is penalized when the data move

Collapse: keep the mixture from growing

After the update each grid point has $2 \times (K+1) = 34$ components; left alone, the count would multiply every year. So, per grid point, the components are merged into two:

- **undisturbed this year** ($k = 0$), and **disturbed this year** ($k \geq 1$)

each replaced by one Gaussian with the same weight, mean and covariance:

$$w = \sum_i w_i^a, \qquad m = \sum_i \frac{w_i^a}{w}\, m_i^a, \qquad C = \sum_i \frac{w_i^a}{w}\left(C_i^a + m_i^a m_i^{a\top}\right) - m\, m^\top$$

This is the best single Gaussian (in KL divergence) for each group. Keeping the disturbed group separate lets next year's forecast still carry a post-disturbance branch.

Summaries

Everything averages over the grid and the components, with weights $\pi_t(g)\, w^a_{g,j,k}$:

$$P(\text{disturbed in year } t \mid y_{1:t}) \;=\; \sum_g \pi_t(g) \sum_j \sum_{k\geq 1} w^a_{g,j,k}$$

$$P\big(x_{t,\text{wood}} \leq u \mid y_{1:t}\big) \;=\; \sum_{g,j,k} \pi_t(g)\, w^a_{g,j,k}\, \Phi\Big(\tfrac{u-m^a_3}{\sqrt{C^a_{33}}}\Big)$$

- 95% credible band: solve for u at 0.025 and 0.975 (root-finding on the mixture CDF) (can be skewed or bimodal, unlike mean $\pm$ 2 sd)
- process error: $\pi_t(g)$ is the posterior of τ; reported as $\sqrt{\mathbb{E}[1/\tau \mid y_{1:t}]}$ or quantiles of $1/\sqrt{\tau}$
- one-step-ahead forecast of y_t: the same mixture before the update; its PIT checks calibration

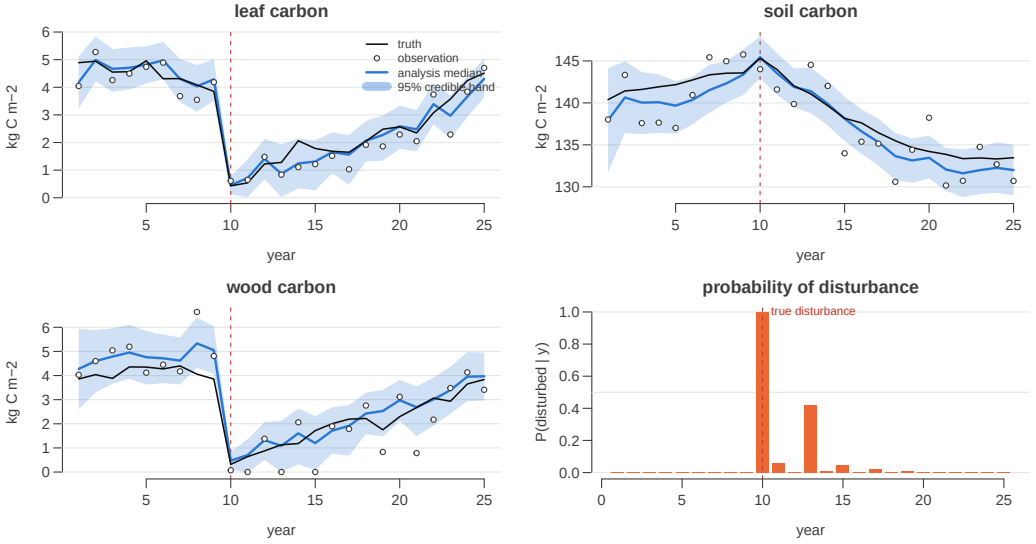
Synthetic test: set-up

The multi-pool version of the filter on simulated data with a known truth:

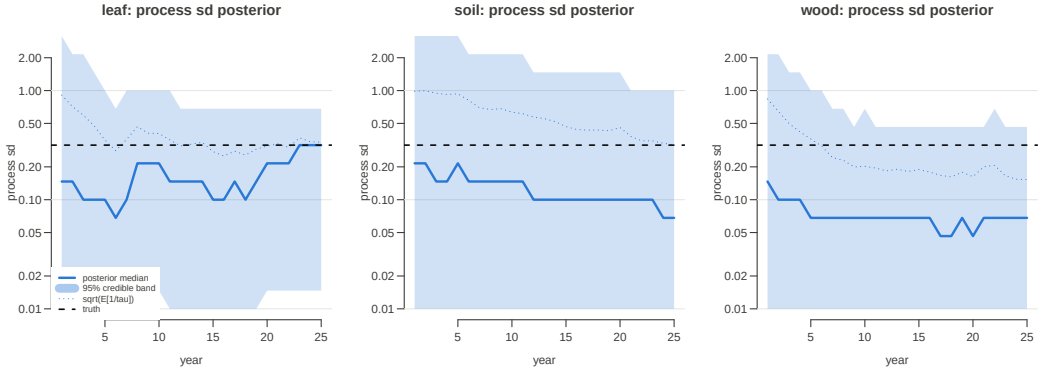
- one site, 25 years, a harvest in year 10 (keeps $\sim$10% of leaf and wood)
- every pool observed, $R = \mathrm{diag}(0.25, 10, 1)$
- true process sd 0.32 for every pool, unknown to the filter: all three precisions are learned, on a 16^3 grid
- prior $P(\text{disturbed}) = 0.05$ every year
- 21 random seeds: new drivers, parameters, truth and observations each time

Differences from the OR model: there only wood is observed and only the wood τ is learned (200-point grid).

Synthetic: states and disturbance (seed 42)

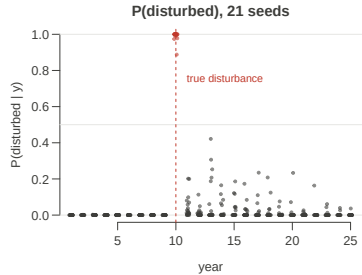


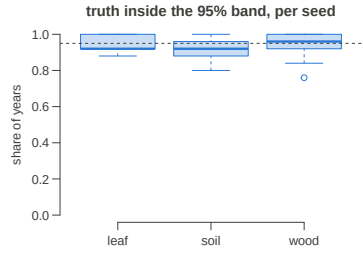
Synthetic: learning the process error (seed 42)



The posterior moves off the log-uniform prior but stays wide after 25 annual observations: the 95% band spans about a factor of 10 and covers the truth, with the median below it and $\sqrt{\mathbb{E}[1/\tau]}$ above it.

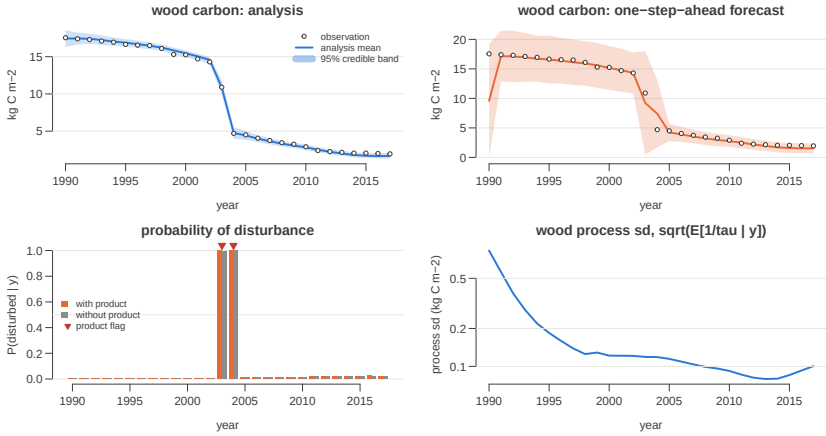
Synthetic: 21 seeds





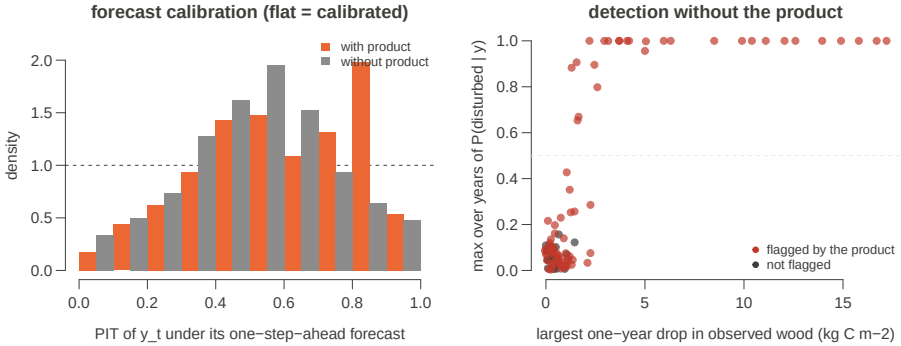
- detection: P(disturbed) in year 10 has median 1.00 (min 0.89); above 0.5 in 21 of 21 seeds
- false alarms ($P > 0.5$ in other years): 0 of 504 seed-years (0 seeds)
- truth inside the 95% band: leaf 94%, soil 93%, wood 94% of years
- RMSE of the median relative to the filter given the true τ: leaf 0.98, soil 1.02, wood 1.01 (medians)
- process sd at year 25: truth inside the 95% band in leaf 16, soil 20, wood 17 of 21 seeds; posterior median / truth leaf 1.47, soil 0.68, wood 1.47 (medians)

OR test run: site 300



GEDI heteroskedastic R, bias correction, gamma prior on τ. Wood falls from 15 to 5 kg C m^{-2} over 2003–04: $P(\text{disturbed}) = 1$ in both years, with or without the product. The first forecast comes from the population-wide initial-condition prior, hence its miss in 1990.

OR: 100 random sites



Left: PIT of each observation after the first under its one-step-ahead forecast. Too many land mid-range (98% in the central 95%, 59% in the central 50%): the forecasts are too wide. Right: without the product, every site with a one-year loss above 3 kg C m^{-2} is detected (20 of 20), none with a loss below 1 (0 of 55). Most sampled sites are product-flagged: the sites were drawn stratified by disturbance class.

OR full grid: set-up and metrics

All 538 sites $\times$ 4 observation-error models $\times$ with / without the disturbance product $= 4304$ runs, with the defaults: bias correction on, process error propagated through the disturbance, gamma prior on τ. About 15 minutes on 8 cores.

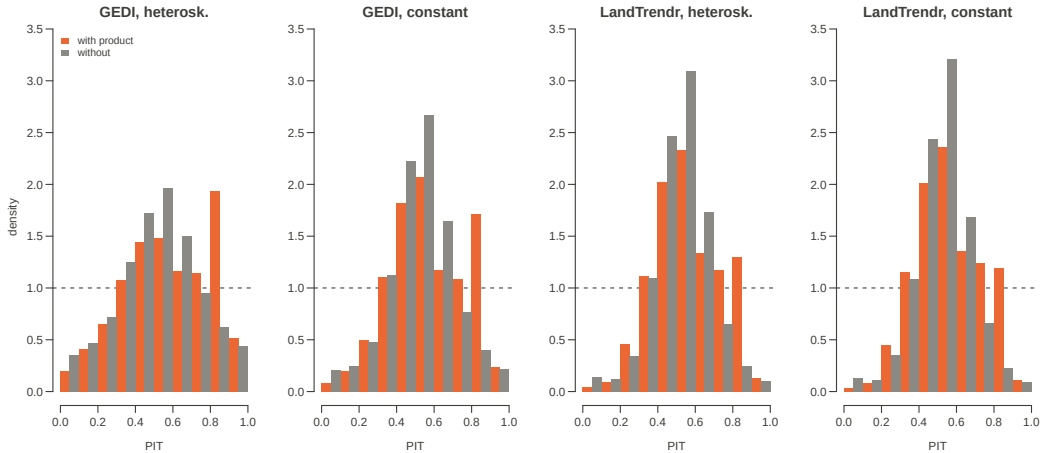
- **PIT in 95% / 50%**: share of observations inside the central 95% / 50% of their one-step-ahead forecast (calibrated: 95% / 50%)
- **final in band**: last observation inside the 95% analysis band
- **detect, loss $>$ 3**: $P(\text{disturbed}) > 0.5$ in years where observed wood fell by more than 3 kg C m^{-2} (151 site-years)
- **detect, flagged**: $P(\text{disturbed}) > 0.5$ in the years the product flags
- **false alarm**: $P(\text{disturbed}) > 0.5$ in unflagged years with $|\text{change}| < 0.5$ (7548 site-years)
- **process sd**: median over sites of $\sqrt{\mathbb{E}[1/\tau \mid y]}$ in 2017 (kg C m^{-2})

OR full grid: results by setting

observation error	product	PIT in 95%	PIT in 50%	final in band	detect loss > 3	detect flagged	false alarm	process sd
GEDI, heterosk.	yes	98%	60%	95%	99%	27%	0%	0.095
GEDI, heterosk.	no	97%	74%	93%	99%	5%	0%	0.111
GEDI, constant	yes	99%	69%	92%	92%	60%	0%	0.131
GEDI, constant	no	98%	84%	92%	89%	3%	0%	0.115
LandTrendr, heterosk.	yes	100%	76%	92%	86%	84%	0%	0.191
LandTrendr, heterosk.	no	99%	90%	92%	68%	3%	0%	0.156
LandTrendr, constant	yes	100%	77%	92%	85%	87%	0%	0.203
LandTrendr, constant	no	99%	90%	93%	60%	2%	0%	0.168

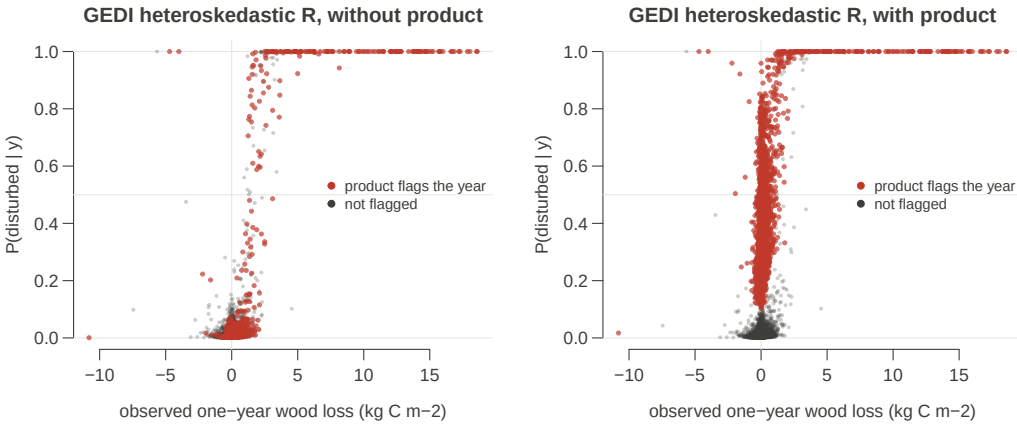
- Forecasts are too wide everywhere (60–90% in the central 50%), most with LandTrendr's larger R.
- Large losses are found without the product under GEDI errors (89–99%); LandTrendr's larger R hides some (60–68%). No false alarms in quiet years.
- Many product-flagged years show little wood loss; under GEDI's small R the data overrule the product.

OR full grid: forecast calibration



PIT of every observation after the first, all 538 sites. A calibrated forecast gives a flat histogram (dashed). The hump means the forecasts are too wide; it is largest with LandTrendr's R.

OR full grid: detection



Every site-year, 1991–2012, GEDI heteroskedastic R. Without the product, P(disturbed) jumps to 1 once the loss passes about 1–2 kg C m^{-2}. With it, flagged years with no visible loss get intermediate probabilities: the prior says disturbed, the data disagree.

Does propagating the process error matter?

Averages over the 8 settings; the earlier grid gave disturbed components no process error:

process error after disturbance	PIT in 95%	PIT in 50%	final in band	detect loss > 3	detect flagged	false alarm	process sd
propagated (this grid)	99%	78%	93%	85%	34%	0%	0.144
none (earlier grid)	99%	77%	93%	85%	34%	0%	0.143

Almost no difference. The added variance, $\mu_0^2/\tau \lesssim 0.02$, is small next to the spread of the disturbance size itself: $V_0 \cdot \text{old}^2$ is roughly 1–30 for a 15 kg C m^{-2} stand.

Summary

- The posterior is a mixture: 2 Gaussians per process-error grid point, plus a weight per grid point.
- Forecast: one batch of VSEM runs, reused for every component by importance weighting; each disturbance type and the process error are applied to it.
- Update: an exact Kalman update per component; the likelihoods reweight the components ($\Rightarrow$ disturbance) and the grid points ($\Rightarrow$ process error).
- Collapse to 2 Gaussians per grid point keeps the cost fixed over time.
- Summaries average over everything: P(disturbed), mixture-quantile credible bands, the posterior of the process sd.
- On the OR grid: large losses are detected without false alarms, but the one-step-ahead forecasts are too wide; propagating the process error through the disturbance changes little.